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SCHOOL OF COMPUTER SCIENCE AND ENGINEERNG

Metadata of your Dataset

1. General Information

1. Dataset Title: **Marine Multimodal Dataset for Cross-Modal Retrieval**

2. Creator(s)/Contributor(s) (Name, designation, contact details) :

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3. Year of Creation / Last Updated:

   Date Created: 23 June 2026, 11:31
   Last Updated: 23 June 2026, 11:50

4. Associated Project / Grant Name (if any): **Symmetric Supervised Contrastive Alignment of Visual, Acoustic, and Semantic Marine Modalities**

5. Purpose / Use-case (e.g., image classification, anomaly detection): **To provide a structured multimodal dataset containing semantically aligned marine mammal images, textual descriptions, and audio recordings for research in multimodal learning. The dataset is intended to support tasks such as cross-modal retrieval, multimodal representation learning, semantic similarity analysis, marine mammal information retrieval, and the evaluation of machine learning models operating across heterogeneous data modalities.**

2. Dataset Description

1. **Brief Abstract / Summary**

This dataset contains semantically aligned marine mammal images, textual descriptions, and audio recordings collected from publicly available and authoritative sources to support multimodal learning research. The dataset includes multiple marine mammal species, with each instance comprising an image, descriptive text, and species-specific audio where available. The data were systematically collected, cleaned, standardized, and organized to ensure semantic consistency across all modalities. The dataset supports cross-modal retrieval, multimodal representation learning, semantic similarity analysis, and biodiversity information retrieval.

2. **Problem Statement or Motivation**

Marine mammal information is distributed across diverse data modalities such as images, textual descriptions, and audio recordings, making efficient access and comprehensive analysis challenging. Existing datasets often focus on a single modality or lack semantic alignment between different modalities, limiting their applicability for multimodal artificial intelligence research. The motivation of this dataset is to provide a unified, semantically aligned collection of marine images, text, and audio that facilitates multimodal representation learning, cross-modal retrieval, semantic understanding, and biodiversity information retrieval while supporting the development and evaluation of advanced multimodal machine learning models.

3. **Description of Features / Attributes**

The dataset contains EEG-derived cognitive and spectral features extracted from wearable ear-EEG signals, including:

Feature Category	**Description**
Audio Dataset	Contains marine mammal species audio recordings organized into train and test directories in standardized audio formats.
Image Dataset	Contains marine mammal species images organized into train and test directories in standardized image formats.
Text Dataset	Contains structured textual descriptions of marine mammal species organized into train and test directories in plain text format.
Dataset Split	Data partitioned into training and testing subsets for model development and evaluation.
Label	Cognitive state category assigned Marine mammal species identifier corresponding to the filename, used to associate related image, text, and audio samples across different modalities.to each sample.

The dataset contains multimodal records comprising marine mammal species images, audio recordings, and structured textual descriptions organized into training and testing subsets, representing **75** distinct marine mammal species across three complementary data modalities.

4. **Target Variable (if supervised):**

   Target Variable: Marine Mammal Species Label

   Type: Multi-class Classification

   Classes: Marine mammal species included in the dataset (e.g., Atlantic Spotted Dolphin, Beluga Whale, Bowhead Whale, Clymene Dolphin, Common Dolphin, False Killer Whale, Fin Whale, Harp Seal, Humpback Whale, Killer Whale, Leopard Seal, Narwhal, Risso's Dolphin and other marine mammal species.)

   The target variable represents the marine mammal species associated with each multimodal sample and is used for supervised machine learning tasks such as species classification, multimodal representation learning, and cross-modal retrieval across image, text, and audio modalities.

3. Data Format and Content

1. **Number of Instances (Rows):** The dataset comprises multimodal samples distributed across image, audio, and text modalities, representing 75 marine mammal species.
2. **Data Types (e.g., numeric, categorical, image, audio, text, time series):**
   - **Image:** Marine mammal photographs and visual captures.
   - **Audio:** Marine mammal vocalizations and underwater acoustic recordings.
   - **Text:** Structured textual descriptions, scientific information, and species-specific documentation.
3. **File Format(s) (CSV, JSON, Excel, Images, Audio, Video, etc.):**
   - **Images:** JPG, JPEG, PNG
   - **Audio:** WAV
   - **Text:** TXT
4. **Data Size (MB/GB): 1.59 GB**
5. **Compression Type (if any): ZIP (.zip)**

Field	**Value**
Number of Instances (Species - Row)	Multimodal samples representing **75 marine mammal species**
Number of Features (Columns)	**Not Applicable** (File-based multimodal dataset)
Data Types	Image, Audio, Text, Categorical

File Format(s)	JPG/JPEG/PNG (Images), WAV (Audio), TXT (Text)
Data Size	1.59 GB
Compression Type	ZIP (.zip)

4. Source and Collection Method

1. **Data Collection Methodology (e.g., sensor-based, scraped, simulated, crowdsourced, captured)**

Web-Based Data Collection and Manual Curation
The dataset was collected from publicly available and authoritative online sources using a combination of automated web extraction and manual curation. Marine mammal images, audio recordings, and textual descriptions were gathered from multiple repositories and scientific resources. The collected data underwent quality screening, deduplication, format standardization, and metadata verification before being organized into modality-specific directories with training and testing subsets. Semantic correspondence between modalities was established based on the associated marine mammal species to facilitate supervised multimodal learning and retrieval tasks.

2. **Data Source(s) (instruments, websites, logs, etc.)**

Primary Data Sources:

- **Image Dataset:**
  Source of data: Primarily sourced from the **Hugging Face dataset yeyimilk/LLM-Vision-Marine-Animals** (https://huggingface.co/datasets/yeyimilk/LLM-Vision-Marine-Animals) and supplemented through manual backend dataset manipulation.
- **Audio Dataset:**
  Source of data: Derived from multi-modal marine repositories and open-source bioacoustic databases, including the **Watkins Marine Mammal Sound Database** (https://huggingface.co/datasets/confit/wmms-parquet) and selected Hugging Face repositories.
- Text Dataset:
  Source of data: Extracted primarily from **Wikipedia** articles corresponding to each marine mammal species (https://www.wikipedia.org/).

3. **Time Period of Data Collection:2025–2026**

5. Licensing and Access

1. Ownership / IP Information
2. License Type (e.g., Open Data Commons, CC-BY, Institutional use only): Open Data Commons
3. Access Restrictions (public, department-only, project-specific): public

6. Quality and Preprocessing

1. Missing Data Information: Only 20 of the 75 marine mammal species have complete image, audio, and text data across all modalities. The remaining species have one or more missing modalities.
2. Cleaning / Preprocessing Done: Duplicate and low-quality samples were removed. Images, audio, and text were cleaned, standardized, and organized into train and test subsets.
3. Data Annotation Details (if applicable): Species labels are assigned using directory names (images/audio) and filename prefixes (text). Cross-modal correspondence is maintained through a standardized naming convention.
4. Validation / Benchmarking Performed: The dataset was validated for file integrity, label consistency, and train/test organization using an 80%–20% split. No model benchmarking has been performed.
5. Known Issues or Limitations: The dataset is limited to marine mammals, with varying sample counts across species and incomplete multimodal coverage for many species.

7. Usage

1. Suggested Applications:
   - Marine Mammal Species Classification
   - Cross-Modal Retrieval
   - Multimodal Representation Learning
   - Marine Mammal Information Retrieval
   - Audio-Visual Species Recognition
   - Bioacoustic Signal Analysis
   - Marine Mammal Acoustic Monitoring
   - Semantic Similarity Learning
   - Multimodal Artificial Intelligence Research
   - Marine Ecology and Conservation Research
2. Existing Publications/Use Cases:
   - **Villon, S., et al. (2018).**
     A Deep Learning Method for Accurate and Fast Identification of Coral Reef Fishes in Underwater Images.
     Use Case: Automated marine species identification and population monitoring using CNN-based image classification.
   - **Qiao, Y., et al. (2023).**
     Underwater Target Detection Based on Improved YOLOv5.
     Use Case: Real-time underwater marine animal detection and localization.
   - **Li, H., et al. (2020).**
     Marine Species Recognition with Image Enhancement and Convolutional Neural Networks.
     Use Case: Marine species classification using enhanced underwater images and CNNs.
   - **Zhong, M., et al. (2020).**
     Automated Classification of Humpback Whale Calls Using Deep Learning.
     Use Case: Automated classification of humpback whale vocalizations from underwater acoustic recordings.

- **Bergler, C., et al. (2019).**
  Deep Learning in Marine Bioacoustics: A Benchmark for Baleen Whale Detection.
  Use Case: Deep learning-based detection of baleen whale calls from passive acoustic recordings.
- **Jiang, W., et al. (2024).**
  Underwater Audio Species Classification Using Dual-Path Deep Learning.
  Use Case: Marine species classification using underwater acoustic signals.
- **Cha, S., et al. (2024).**
  Leveraging Large Language Models for Processing Unstructured Citizen Science Marine Data.
  Use Case: Extraction of structured marine species information from unstructured textual data.

8. Recommended Models / Baselines (if any):

Recommended Deep Learning Models

Image Models:

- ConvNeXt V2
- DINOv2

Text Models:

- MPNet
- BERT

Audio Models:

- Audio Spectrogram Transformer (AST)
- Wav2Vec 2.0

9. Storage and Sharing

1. Storage Location / Link (Google Drive, Git, Internal server): https://ieee-dataport.org/documents/marine-dataset-1
2. Versioning (if multiple versions exist):NA
3. Contact Person for Access or Clarification: Hariprasath M

Place : Chennai

Signature of the Data Owner

Date : 26-06-2026

Associate Dean (Research)

Dean SCOPE